



AARJAN BUDATHOKI

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WORK EXPERIENCE

Embedded Firmware Developer

Jul 2025 – Present

Yatri Motorcycles Pvt. Ltd.

Budhanilkantha, Nepal

- Developed embedded firmware for STM32 and ESP32-based automotive VCU using C/C++ for the Yatri P2 electric scooter, contributing to one of Nepal's first locally manufacturing EV company.
- Integrated the Fardriver motor controller with the P2 VCU over CAN, implementing hill hold, cruise control, speed limiting, regenerative braking, controller fault monitoring, and system-wide error reporting for diagnostics and logging.
- Developed firmware for an STM32H743 MCU, Lawicel SLCAN-compatible USB-to-CAN adapter.
- Designed and developed a cross-platform (Windows/Linux) GUI application for automated instrument cluster testing, supporting CANalyst-II, PEAK CAN, and SLCAN-compatible USB-to-CAN adapters.
- Developed software-based mock devices for Hardware-in-the-Loop (HIL) testing to simulate battery, motor controller, display, onboard charger, and handlebar communication, enabling feature validation without complete vehicle hardware.
- Implemented an OTA update framework on ESP32-S3 with firmware download, TAR packaging, TinyUZ decompression, and HPatchLite-based delta OTA support for VCU and Allwinner-based display devices.

Sensor Fusion and Perception Intern

Dec 2024 - Mar 2025

LogicTronix [FPGA Design & Machine Learning Company]

Lalitpur, Nepal

- Developed robust pipelines for data acquisition & visualization of Velodyne 3D LiDAR sensor data using the Point Cloud Library (PCL).
- Annotated custom 3D point cloud datasets using 3D bounding box labeling tools(Latte).
- Implemented voxel grid down-sampling algorithms to optimized point cloud density for computationally efficient processing.
- Implemented RANSAC-based ground plane segmentation in C++ to effectively isolate dynamic objects such as pedestrians and vehicles from static ground surfaces.
- Implemented DBSCAN clustering algorithm in C++ for object detection and classification, targeting pedestrians, vehicles and traffic objects.
- Developed and optimized Kd-Tree data structure from scratch in C++ to accelerate nearest-neighbor search operations on large-scale point clouds.

Electronics Hardware Intern

Nov 2023 – Feb 2024

Abhiyan Engineering Nepal Pvt. Ltd.

Lalitpur, Nepal

- Contributed to hardware system development for the Automatic Sand Level Measurement Embedded System (PSLM Device), focusing on sensor integration, circuit design, and system prototyping.

Work Student

Aug 2022 – Sep 2023

Robotics Club, IOE Pulchowk Campus

Lalitpur, Nepal

- Implemented kinematics modeling and odometry of four omniwheeled motion robot.
- Implemented PID controllers for position and velocity tuning of brushed and brushless DC motor.
- Manual tuning, heuristic based tuning called ziegler nichols tuning and fuzzy control systems.
- Design & Fabrication of main controller board of the R2 BOT and power supply of bots in ABU ROBOCON 2022.

PROJECTS

Greenhouse Management using AWS IoT and Yocto Embedded Linux

2025

Major Project, Tribhuvan University

- Embedded Linux: Developed an Embedded Linux OS for Raspberry Pi 4 (64-bit) using Yocto Project, featuring tools like git, python3, opkg, connman, mosquito, and systemctl. Enabled UART, modifiable root filesystem, and WIC image support for deployment.

Self-Driving Car

Mar 2024

Minor Project, Tribhuvan University

- Integrated `ros2_control` for precise control of car with Ackermann Wheel Steering.
- Implemented SLAM algorithms to generate accurate Lidar-based maps of static environments.
- Incorporated `nav2` stack for autonomous navigation and real-time path planning.

Elephant and Rabbit Robot

Aug 2023

ABU Robocon 2023 (Cambodia), Robotics Club Pulchowk Campus

- Used the scikit-learn framework DBSCAN clustering algorithm and Nearest Neighbors library for detecting field poles using 2D LiDAR point cloud data to accurately identify and localize the nearest pole within the game environment.

R1 and R2 Robot

Aug 2022

ABU Robocon 2022 (India), Robotics Club Pulchowk Campus

Maze Generation and Solving Algorithm

Feb 2023

Data Structures and Algorithms Project, 4th Semester

- * Implemented Depth-First Search (DFS) algorithm for procedural maze generation.
- * Applied shortest pathfinding algorithms, including Dijkstra's Algorithm and A* Search, for efficient maze solving.

Self-Balancing Robot

Jul 2022

Instrumentation Project 3rd Semester

- * Implemented PID Control for Angle Equilibrium using IMU.

AI/ML RELATED PROJECTS

Traffic Sign Detection using CNN

Preprocessed a labeled traffic sign dataset by resizing, normalizing, and augmenting images to improve model robustness. Built and trained a CNN model using tensorflow and keras, for accurate classification and detection of traffic signs.

End to End Learning for Self Driving Cars in Udacity Simulator

Designed and trained an Nvidia CNN model using TensorFlow for behavior cloning in a self-driving car simulator. Improved model robustness through data augmentation techniques, including image flipping, brightness adjustment, and translation.

EDUCATION

Bachelor in Electronics, Communication and Information Engineering

Tribhuvan University, Institute of Engineering, Pulchowk Campus

Apr 2021 – Present

Lalitpur, Nepal

HONORS AND AWARDS

Mabuchi Motor Award ABU Robocon 2023, Asia-Pacific Robotics Competition, held in Cambodia	Aug 2023
2nd Runner Up ABU Robocon 2022, Asia-Pacific Robotics Competition, held in India	Aug 2022
Nagase Award ABU Robocon 2022, Asia-Pacific Robotics Competition, held in India	Aug 2022
Dronacharya Title Winner LOCUS 2024	Jan 2024

TEACHING EXPERIENCE

Hardware Fellowship 2024 Robotics Club & Locus 2025, 8 Days Workshop	Jul 2024 IOE, Pulchowk Campus
Robo Line Dash Locus 2024, 2 Days Workshop	Jan 2024 IOE, Pulchowk Campus

COMMUNITY INVOLVEMENT

Robotics Club, Pulchowk Campus Participation in ABU Robocon 2023	Apr 2022 – Present Lalitpur, Nepal
IEEE Student, Pulchowk Campus General Member	Jan 2024 – Present Lalitpur, Nepal
LOCUS 2025, Pulchowk Campus Executive Committee Member(Hardware Coordinator)	Jun 2024 – Present Lalitpur, Nepal

SKILLS

Programming: C, C++, Python
Embedded: STM32, ESP32, FreeRTOS, STM32Cube Suite
Robotics: ROS 2 (ros2_control, Nav2, SLAM), PCL, OpenCV, YOLOv8
Tools: Git, GDB, Oscilloscope, Logic Analyzer, KiCad, VeloView, Latte (3D Annotation)
Hardware: Velodyne VLP-16 3D LiDAR